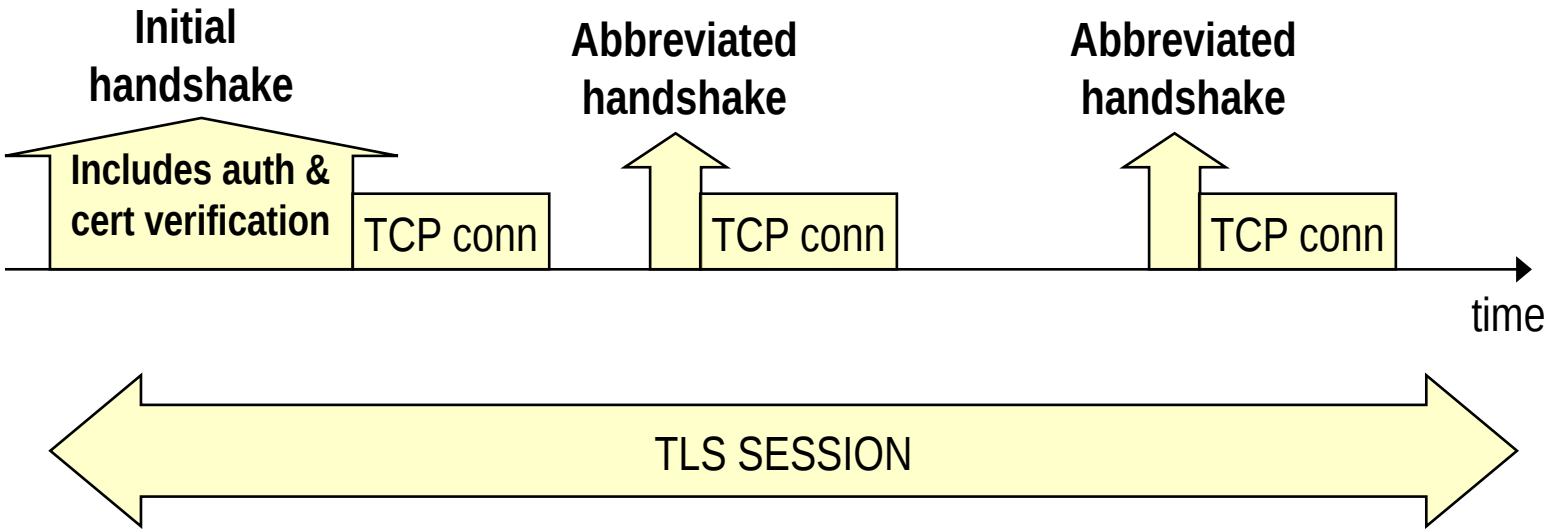
TLS Handshake Protocol

up to TLSv1.2!!!
TLSv1.3 differs, more later

WARNING: content of these slides is LARGELY inferior than content of lectures!
If you don't attend lectures remember to fill gaps…

Handshake: when

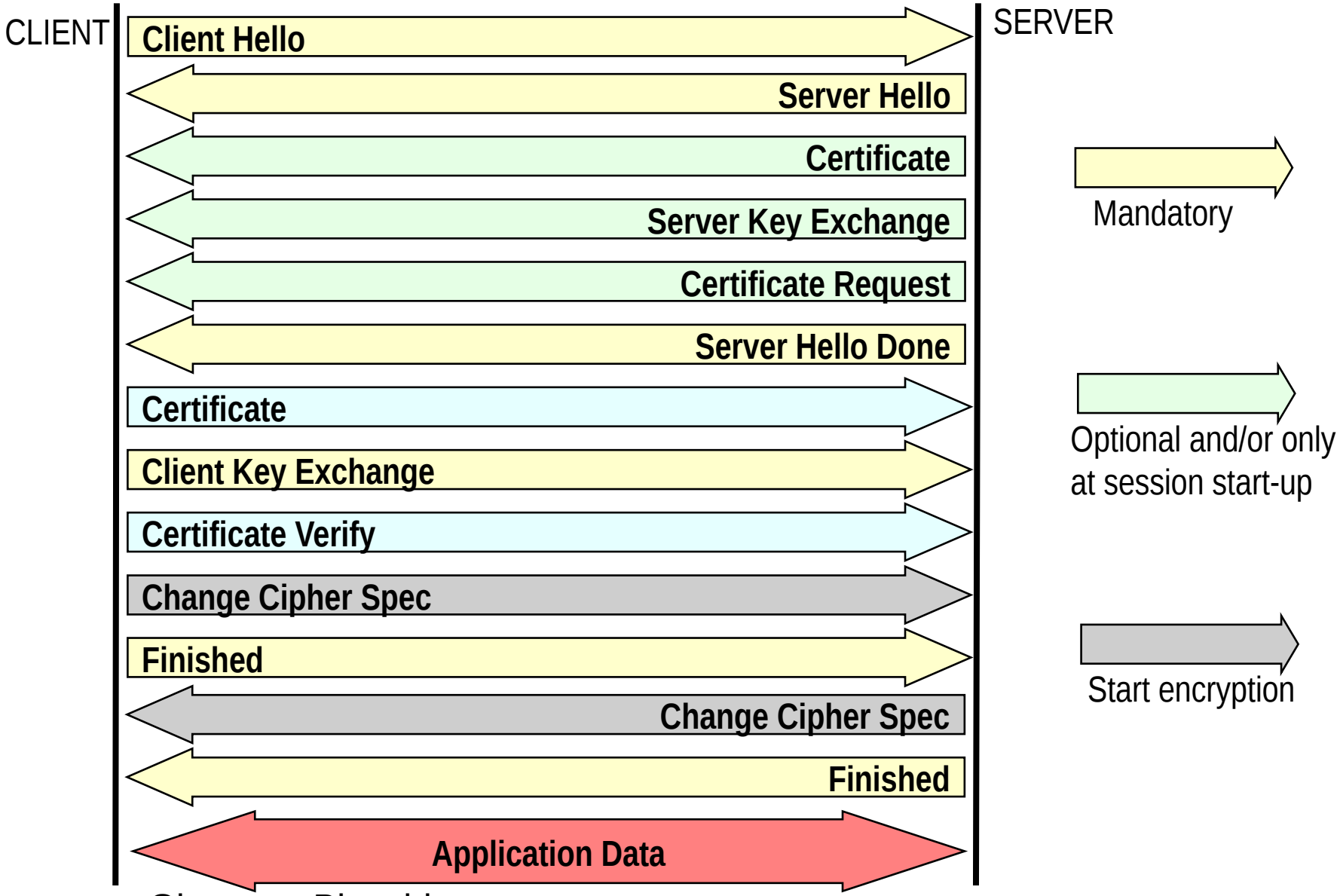
- **Initial negotiation and exchange of parameters**
  - (mutually) authenticate
  - agree on algorithms
  - exchange random values
  - Exchange secrets or information to compute secrets
- **session resumption**
  - Re-keying (“refresh” secrets)



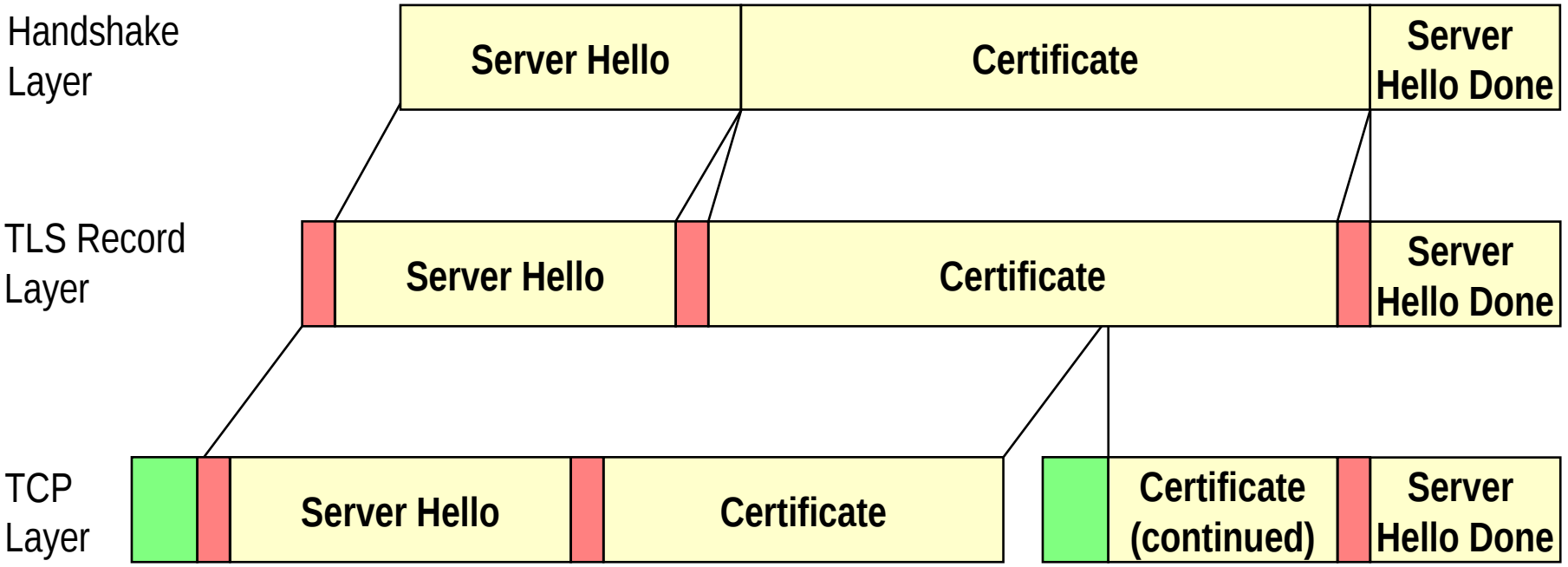
Handshake: goals

- **Secure negotiation of shared secrets**
  - **Via Asymmetric cryptography**
  - Never transmitted in clear text;
  - derived from crypto parameters exchanged
- **Optional authentication**
  - for both client and/or server
    - in practice always required for server
  - Robust to MITM attacks
- **Reliable Negotiation:**
  - Attacker cannot tamper communication and affect/alter the negotiation outcome without being detected by the involved C&S parties
    - Fundamental: negotiation amenable to downgrade attacks!!

Handshake Messages



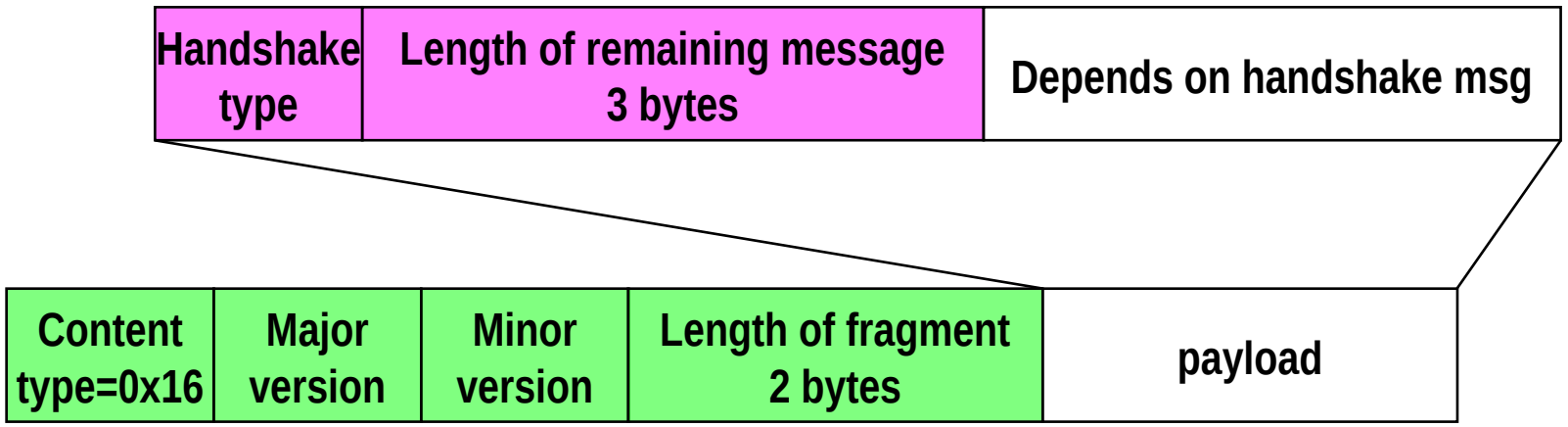
TCP segmentation (example)



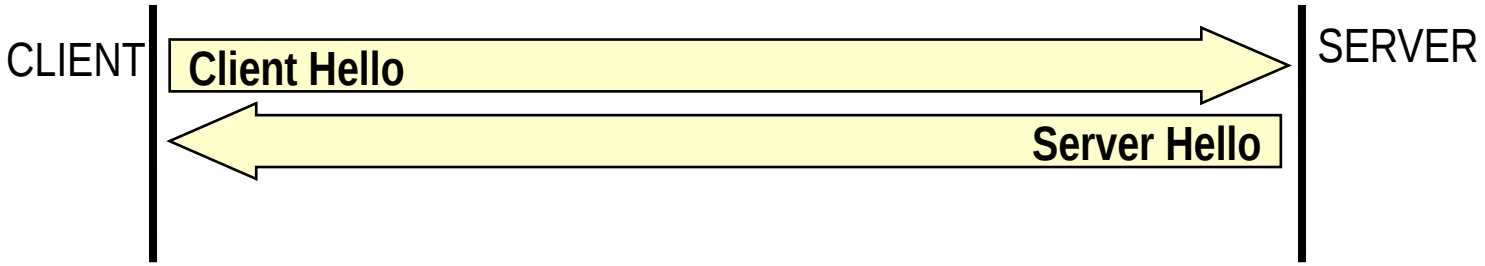
- **Simplified example: server responds to client hello with certificate only**
- **Arbitrary correspondence between TLS Records and TCP segments (TLS records fragmented into TCP segments)**
  - Obvious! Do you remember TCP? ☺

Handshake message format

- **Incapsulated in TLS Record**
- **Handshake types:**
  - hello_request(0), client_hello(1), server_hello(2),
  - certificate(11), server_key_exchange (12),
  - certificate_request(13), server_hello_done(14),
  - certificate_verify(15), client_key_exchange(16),
  - finished(20)



Handshake phase 1



- **General goal:**
  - Create TLS/SSL connection between client and server
- **Detailed goals**
  - Agree on TLS/SSL version
  - Define session ID
  - Exchange nonces
    - timestamp + random values used to, e.g., generate keys
  - Agree on cipher algorithms
  - Agree on compression algorithm

Client Hello

- **First message, sent in plain text**
- **Incapsulated in 5 bytes TLS Record**
- **Content:**
  - Handshake Type (1 byte), Length (3 bytes), Version (2 bytes)
    - Type 01 for Client Hello
    - Version: 0300 for SSLv3, 0301 for TLS
  - 32 bytes Random (4 bytes Timestamp + 28 bytes random)
    - First 4 bytes in standard UNIX 32-bit format
      - Seconds since the midnight starting Jan 1, 1970, GMT
  - (Session ID length, session ID)
    - 1+32 bytes – either empty or previous session ID
      - Previous session ID = resumed session state
  - (Cipher Suites length, Cipher suites)
    - 2+Nx2 bytes, in order of client preference
  - (Compression length, compression algorithm)
    - 1+1 byte (in all cases, today: compression algo = 00 = null)

Cipher suites

PUBLIC-KEY ALGORITHM | SYMMETRIC ALGORITHM | HASH ALGORITHM

TLS_AAAAAAA_WITH_BBBBBBBB_CCCCCCCC

TLS_NULL_WITH_NULL_NULL ← INITIAL (NULL) CIPHER SUITE

TLS_RSA_WITH_NULL_MD5
TLS_RSA_WITH_NULL_SHA

TLS_RSA_WITH_RC4_128_MD5
TLS_RSA_EXPORT_WITH_RC2_CBC_40_MD5
TLS_RSA_WITH_3DES_EDE_CBC_SHA

TLS_DH_DSS_WITH_3DES_EDE_CBC_SHA
TLS_DH_RSA_WITH_DES_CBC_SHA

TLS_DHE_DSS_WITH_DES_CBC_SHA
TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA

TLS_DH_anon_WITH_RC4_128_MD5

JUST A FEW EXAMPLES
(FULL LIST IN RFC)

Note: arbitrary combination not possible: must choose from a (small) list of combinations

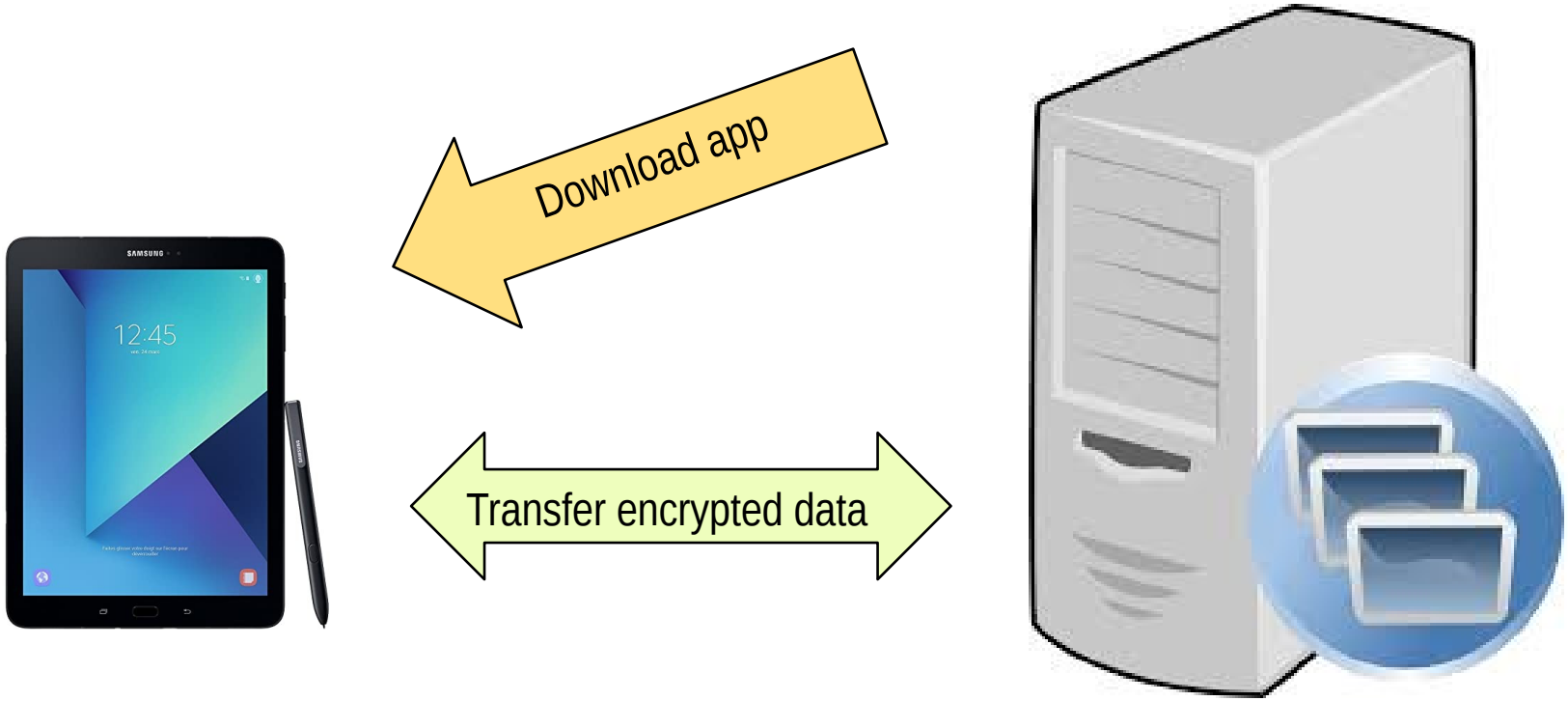
Server Hello

- **In reply to Client Hello, sent in plain text**
- **Content:**
  - Handshake Type (1 byte), Length (3 bytes)
    - Type = 02 for Server Hello)
  - Version
    - Highest supported by both Client and Server
  - 32 bytes Random (4 bytes Timestamp + 28 bytes random)
    - Different random value than Client Hello
  - (Session ID length, session ID)
    - If Client session ID=0, then generate session ID
    - Otherwise, if resumed session ID OK also for Server, use it;
    - Otherwise generate new one
  - Cipher Suite, 2 bytes
    - Selected from client list (usually best one, but no obligation)
  - Compression algo, 1 byte
    - Selected from Client list

Interlude: how public key cryptography is used in TLS

(basic idea, details later on)

Problems solved by asymmetric Encryption: a typical example



How to transfer and store symmetric key?

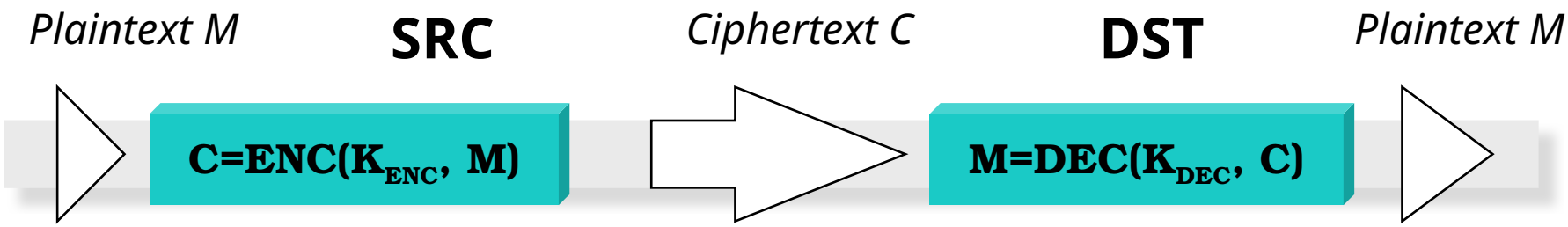
- **In the code? Not nearly a good idea!!!**
- **But then, how to?**

Note the difference with cellular → key in the SIM

Asymmetric cryptography

M plaintext

C ciphertext



K_{ENC} K_{DEC}

K_{ENC} encryption key

ENC(.) encryption algorithm

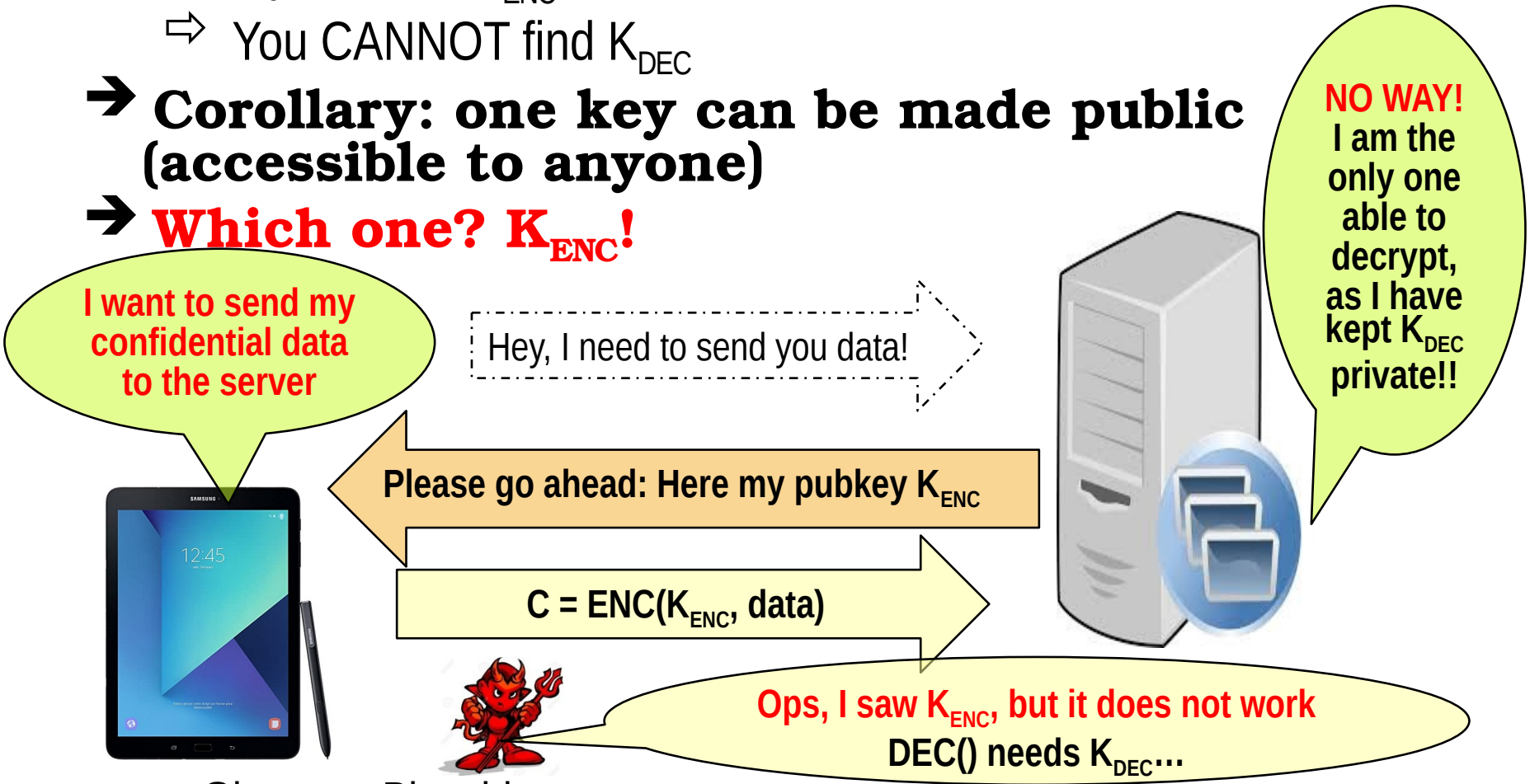
$K_{DEC} \neq K_{ENC}$ decryption key

DEC(.) decryption algorithm

Encryption and decryption keys are obviously linked…
But CANNOT be determined from each other
(unless you know 'more')

How to encrypt data? Public and private key!

- **Asymmetric key property:**
  - If you know K_{ENC} …
  - You CANNOT find K_{DEC}
- **Corollary: one key can be made public (accessible to anyone)**
- **Which one? K_{ENC}!**



Asymmetric vs Symmetric?

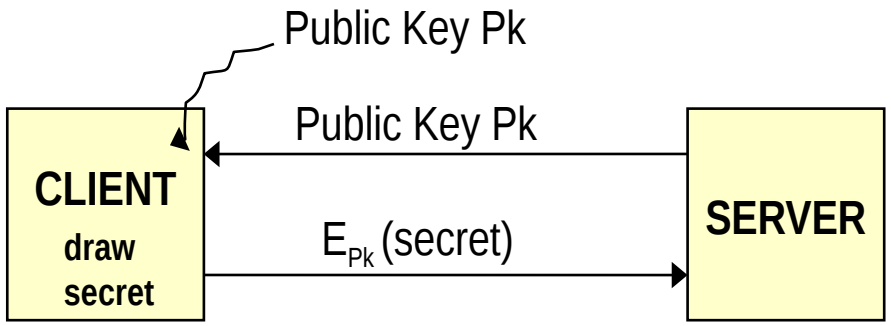
- **They do address DIFFERENT problems**
  - Would be wrong to say that one is superior than the other!
- **BOTH can be (in)secure**
  - Depends on algorithms (RSA, El Gamal, etc)
  - Depends on key sizes
- **Asymmetric is certainly more flexible…**
  - No need to preshare keys – more later
  - But you need also more complex protocols!
- **But asymmetric is computationally heavier than symmetric**
  - 4-5 orders of magnitude slower! Harder to accelerate

Take-home: use asymmetryc crypto to exchange (or setup) a shared key and THEN transfer massive data using symmetric encryption

Actually, TWO basic approaches to key management (both possible until TLS1.2)

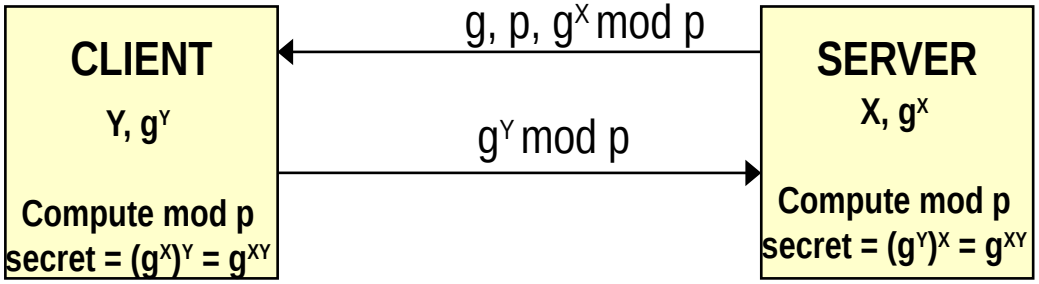
➔ 1. Key transport (e.g., RSA)

- ⇨ Server sends Public Key
- ⇨ Client draws random secret
- ⇨ Which is encrypted using server Pk
- ⇨ And transported to the server
- ⇨ **Both peers now have the same secret**



➔ 2. Key agreement (e.g., DH)

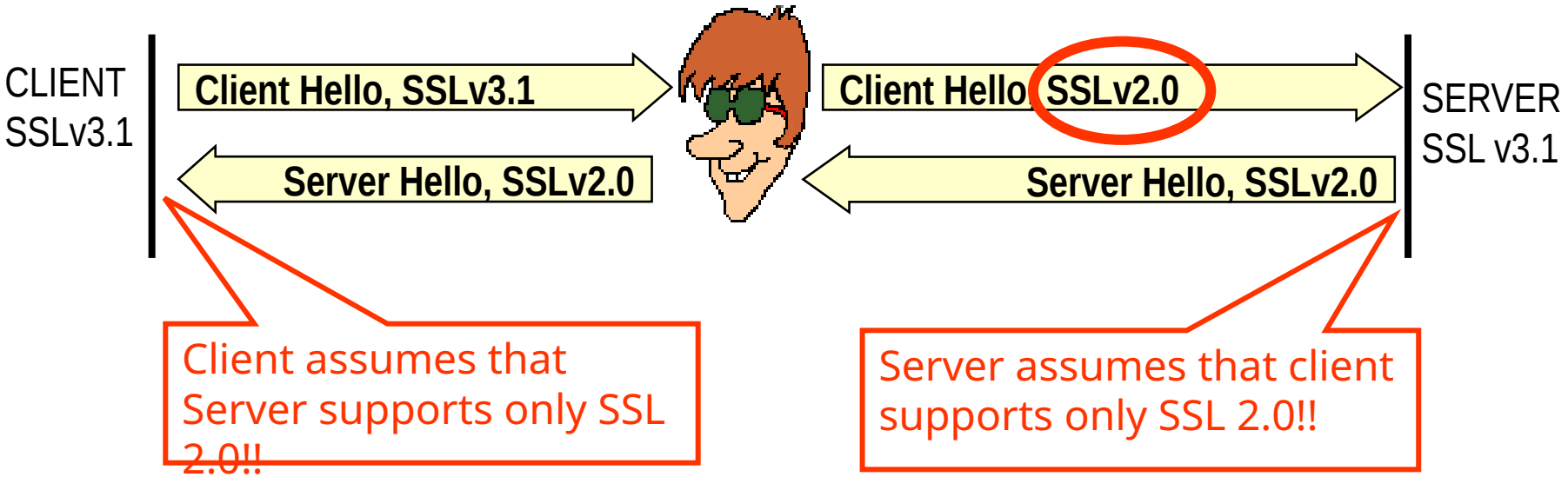
- ⇨ Parties independently generate a private and a public quantiity
- ⇨ Then exchange ONLY the public quantity
- ⇨ This, combined with the other party's private part, permits both of them to compute a SAME SECRET
- ⇨ **Both peers now have the same secret**



Back to TLS handshake...

Downgrade attack

MITM – doesn't break SSL3.x, but knows very well how to break SSL2.0 (40 bit encryption only)

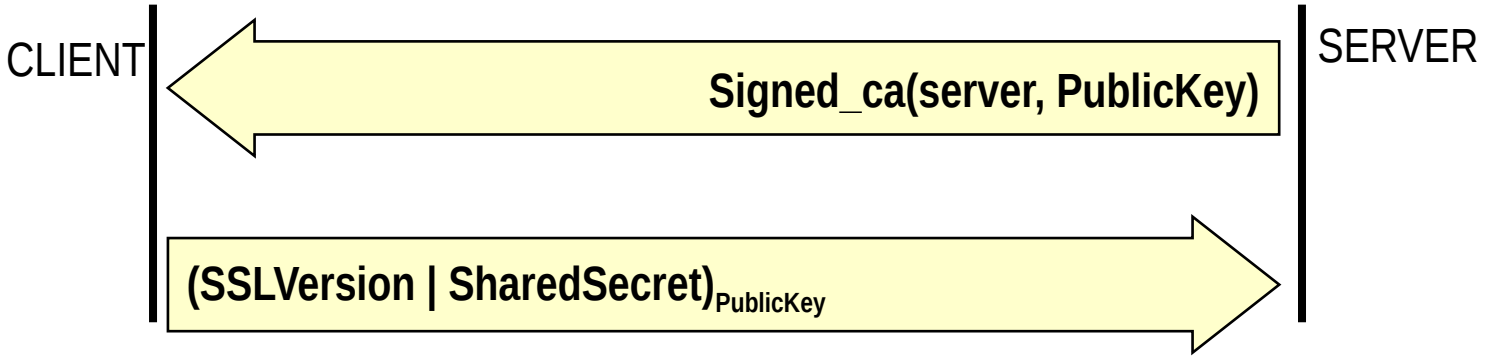


➔ **Downgrade attacks on cipher suites, too**

- ⇨ MITM: remove "strong" cipher suites, and leave in Client Hello only the ones he knows he can break!

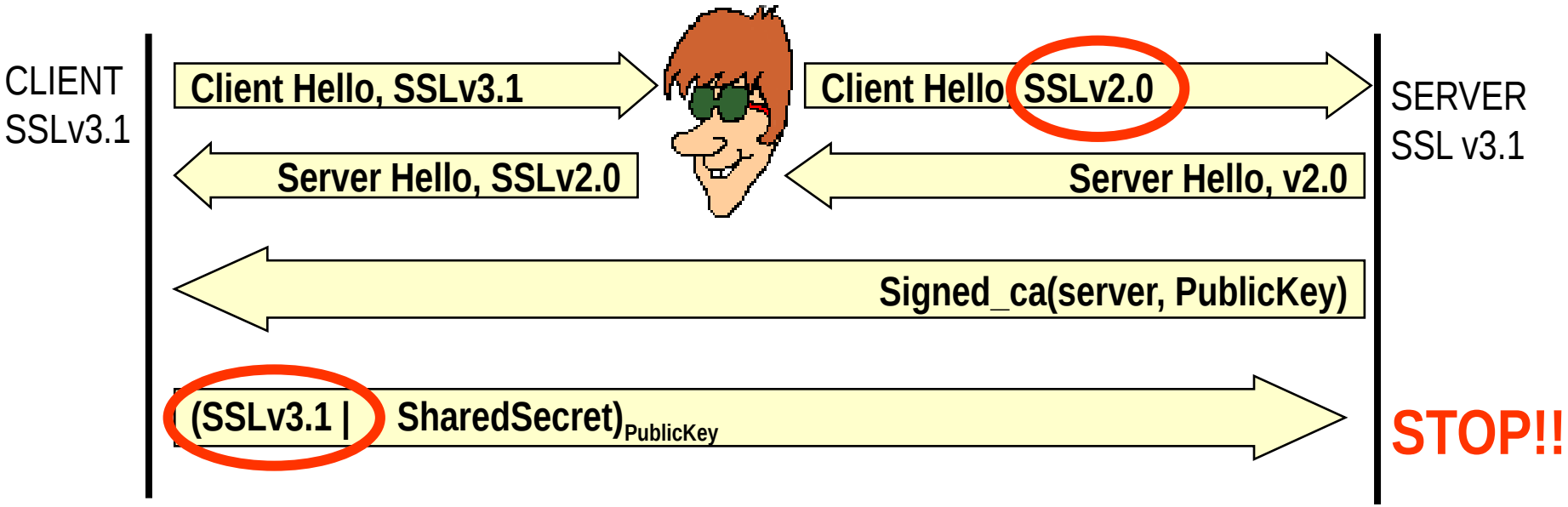
Since hello message authentication not viable at the moment (not yet a shared secret available), verification must be necessarily delayed to a subsequent phase…

Handshake phase 2 & 3 (schematic, RSA case)



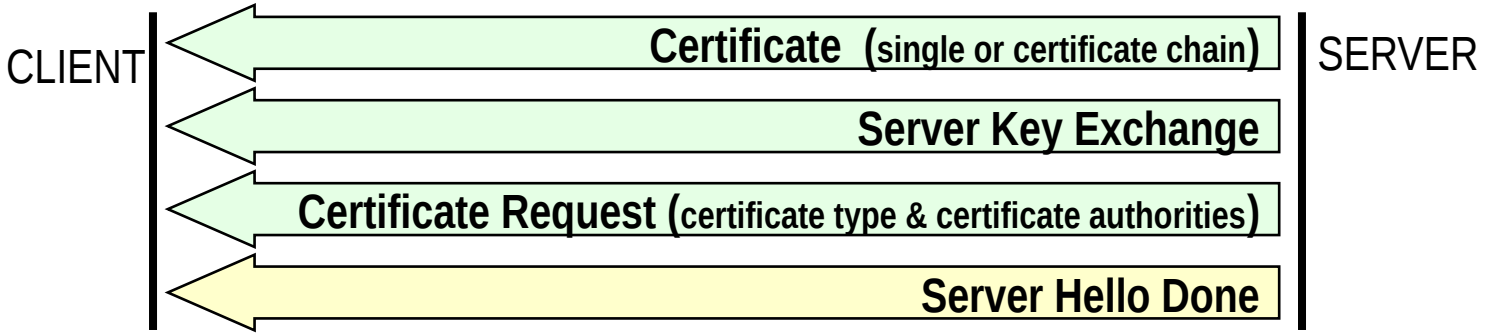
- **Phase 2:**
  - Server sends authentication information (certificate)
  - Along with Public Key (in certificate or in extra msg)
- **Phase 3:**
  - Client generates a Shared Secret
    - Length depends on agreed cipher suite (RSA = 46 bytes)
  - And transmits it to Server, encrypted with Public Key
    - Native SSL version included to early combat downgrade attacks

Against downgrade attack



- **Since (signed) certificate cannot be tampered…**
- **… and since MITM cannot decrypt SharedSecret**
  - and must act as pass-through
- **Downgrade attack on SSL version easily discovered!**
  - Encrypted SSL version is the one initially proposed, NOT the one negotiated in server hello, of course

Handshake Phase 2 details



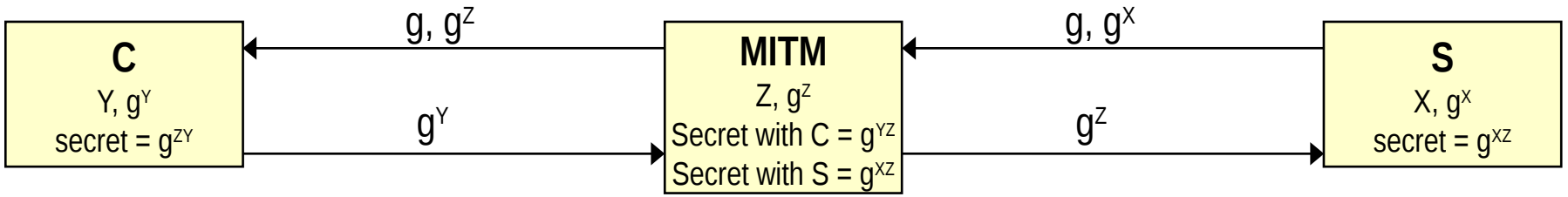
- **Certificate**
  - typically a certificate chain
- **Server Key exchange**
  - When certificate not issued (no server authentication required – UNSAFE!!)
  - When certificate key is for signature only (not for encryption)
  - When certificate key cannot be used for legal reasons
    - E.g. RSA key larger than 512 bits may not be used for encryption outside US, but only for signature
    - Larger RSA key encoded in certificates may be used to sign shorter, temporary, RSA key
  - When Ephemeral Diffie-Hellman used
- **Certificate request**
  - Requires client authentication (asks for an acceptable certificate)
- **Server Hello Done**
  - empty message

Example: Diffie-Hellman /1

- **Review of DH**
  - Server transmits public value g^X mod p
  - Client replies with public value g^Y mod p
  - Both compute shared key as
    - $K = (g^Y)^X \bmod p = (g^X)^Y \bmod p$
- **Three basic approaches (see next slide):**
    - Anonymous
    - Fixed
    - Ephemeral
  - Depends on whether X and Y values are pre-assigned or dynamically generated and/or signed
  - Anon & Ephemeral DH transmitted in ServerKeyExchange

Example: Diffie-Hellman /2
DH variants

- **"Anonymous" (basic) Diffie-Hellman**
  - X, Y generated on the fly and NOT authenticated
  - Trivial MITM attack

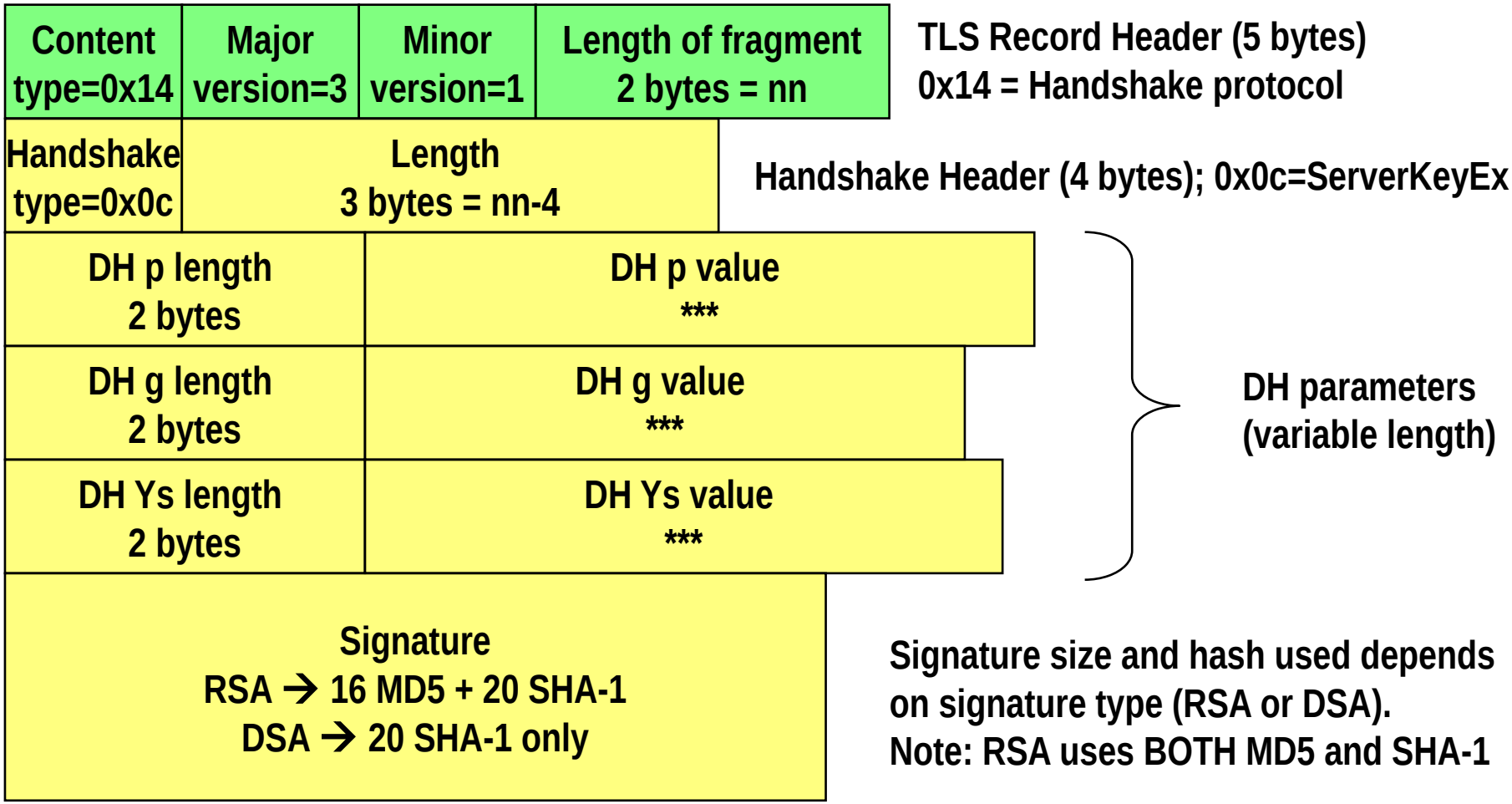


- **(Fixed) Diffie-Hellman**
  - DH "public" parameters g^X and g^Y are static and signed by a certification authority
  - No MITM anymore
  - But they are long-lived → brute-force attacks

- **Ephemeral Diffie-Hellman**
  - DH parameters can vary
  - But unlike anonymous DH, they are SIGNED by the party
  - Hence parties mush have an RSA or DSS secret key for signature purposes
    - → Hence, why don't just use RSA key transport?
      (we are here neglecting IPR/legal issues, of course ☺)

Example: Diffie-Hellman /3

ServerKeyExchange message format:



Client replies with DH public value Ys only (p and g are already known by the Server ☺)

Interlude: Entity authentication with asymmetric crypto

informal sketch, tricky details in Menezes, chapter 10.2

Entity Authentication in a pubkey setting

➔**WRONG:**

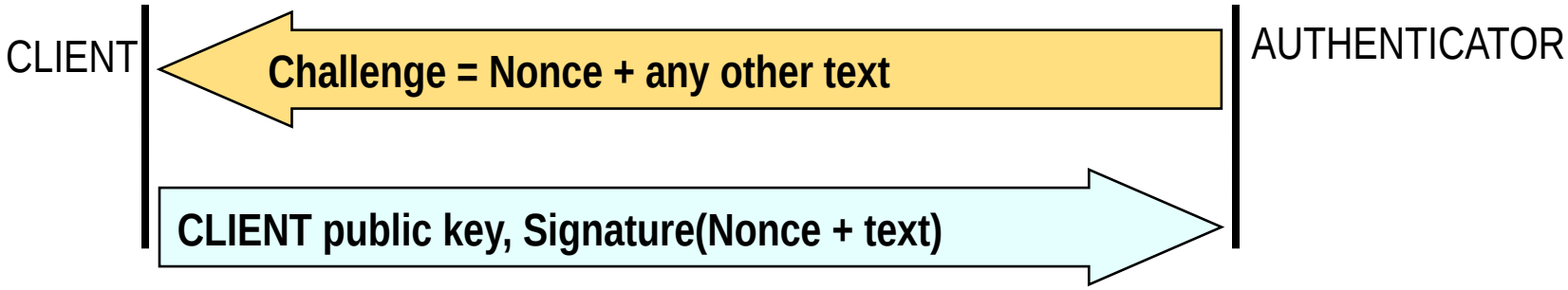
- ⇨ Show your certificate
- ⇨ Why? ☺

➔**RIGHT:**

- ⇨ Prove you know the private key associated to your public key

➔**How to prove such knowledge?**

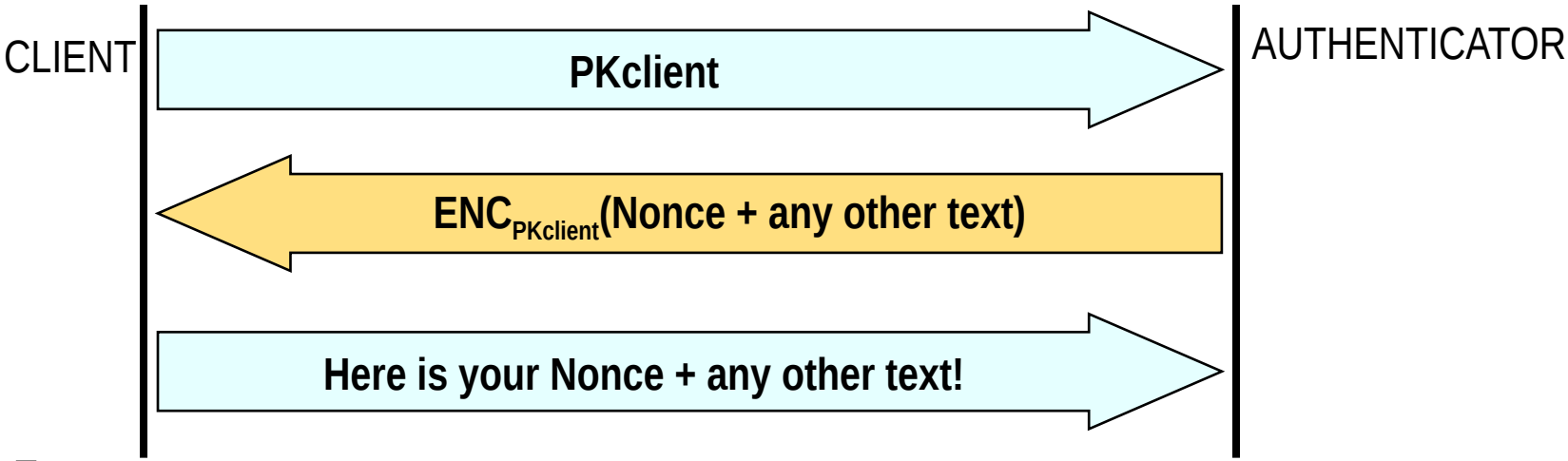
1) Use digital signature



➔Rationale:

- ⇨ I can sign only if I own the private key corresponding to a given public key
- ⇨ of course, pubkey must be bound to client identity → certificate

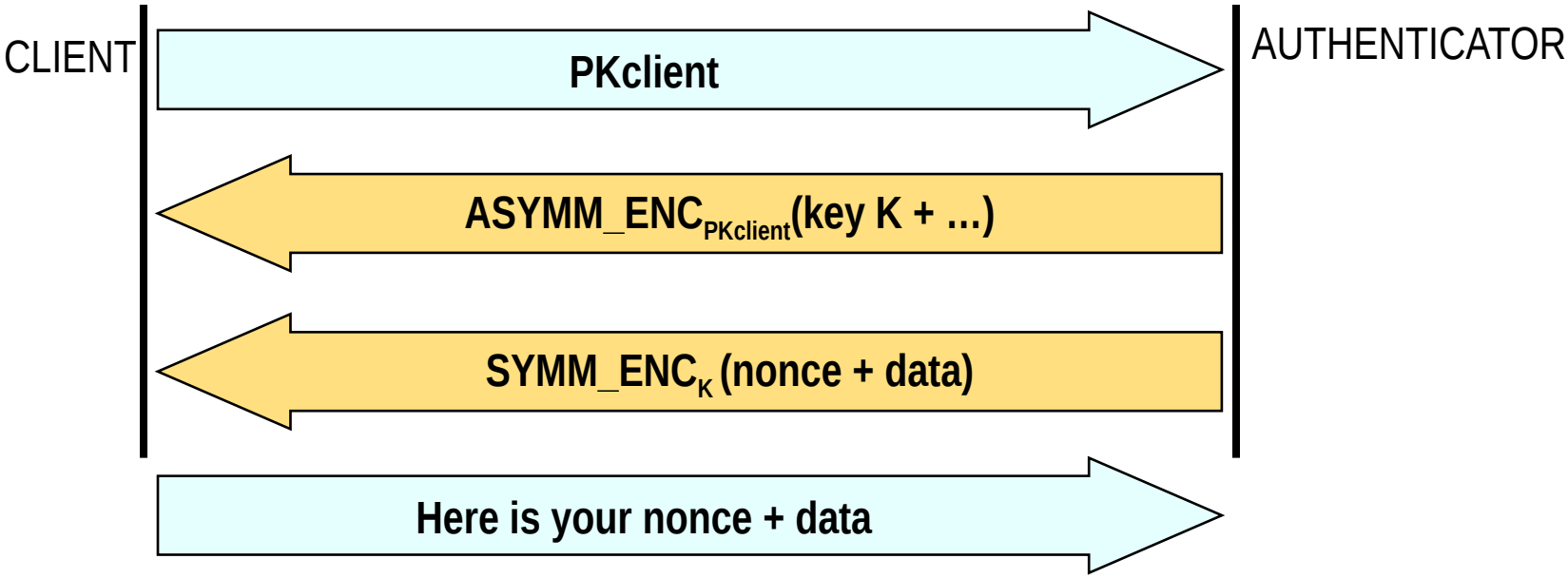
2) Use encryption
(prove you can decrypt!)



➔ **Rationale:**

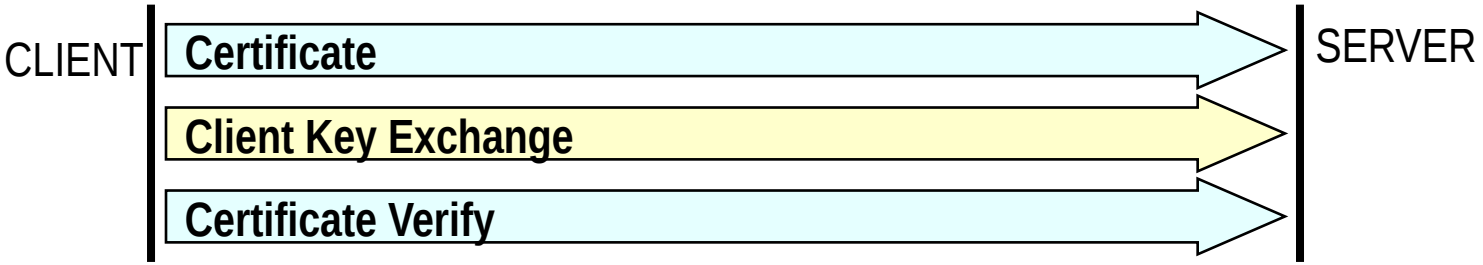
- ⇨ I can decrypt only if I own the private key corresponding to **the public key you used to encrypt!**
- ⇨ Same as before: pubkey = certificate

2bis) prove knowledge of transferred key



Back to TLS!
Entity authentication with asymmetric crypto

Handshake Phase 3 details

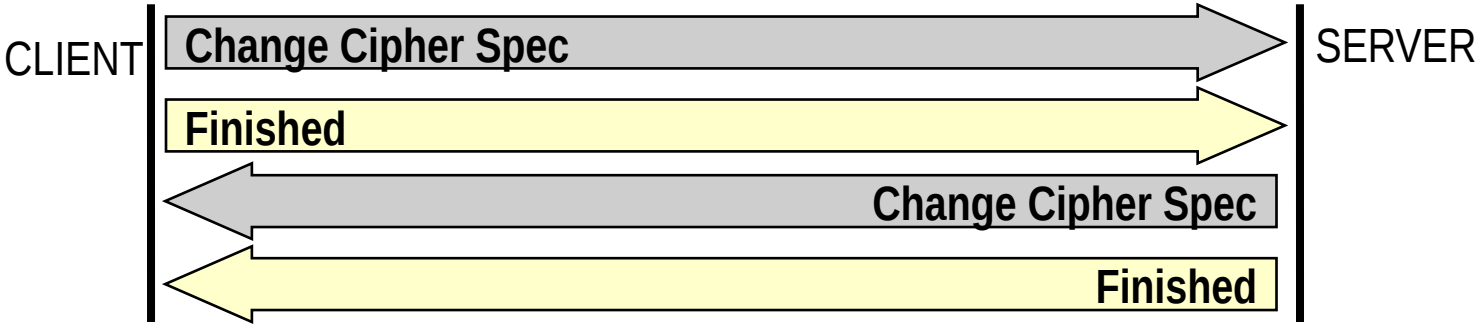


- **Certificate:**
  - Client certificate if requested
- **Client Key exchange**
  - Transmit encrypted symmetric (premaster) key or information to generate secret key at server side (e.g. Diffie-Hellman Ys)
- **Certificate Verify**
  - Signature of "something" known at both client and server
    - ALL the messages exchanged up to now
      - Which is not only known, but also useful! Allows to detect at an early stage (more later on this) tampering attacks (e.g. cipher suite downgrade)
  - To prove Client KNOWS the private key behind the certificate
    - Otherwise I could authenticate by simply copying a certificate ☺

A (smart) detail on certificate verify

- **Q: Why Certificate Verify does not immediately follows certificate?**
- **A: to include connection specific crypto parameters into signature!**
  - client & server random values
  - Explicit inclusion of master secret in SSLv3.0
    - Since master secret can be computed only after the ClientKeyExchange message, Certificate Verify follows ClientKeyExchange
    - Although this was abandoned in TLS1.x, but certificate verify still left as late as possible and hence after ClientKeyExchange

Phase 4



- **Phase 4 has two fundamental goals**
  - Switch to the new security connection state
    - Hence authentication and encryption based on keys computed with the exchanged security parameters
    - Immediately applied to finished message
      - First check that everything went OK! If Client and(orserver cannot decrypt finished message, something has gone wrong!
  - Authenticate all the previous handshake messages
    - Finished = digital signature of all the previous handshake messages as transmitted and received by the peer up to now
    - To avoid MITM tampering

- **Server authentication? Here as well!!!**

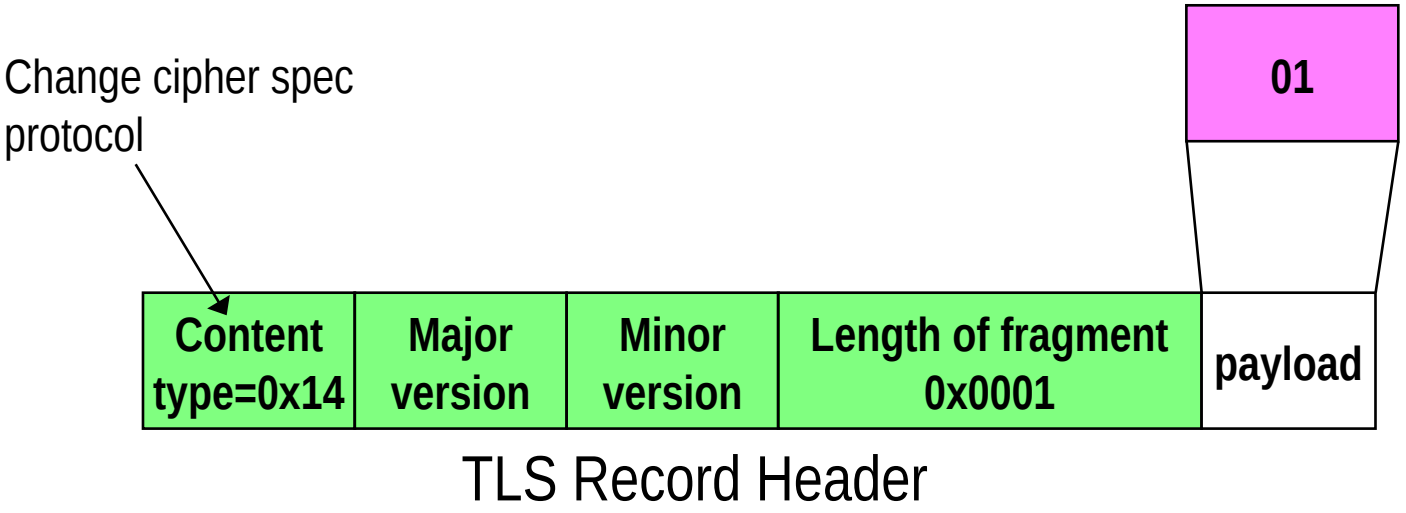
What if negotiation = encrypt but no auth?

➔GAME OVER!

➔Why? ☺

Change Cipher Spec

- **Defined as a separate protocol (!)**
- **Only one message:**
  - 1 single byte
  - Fixed content: constant value = 01 (!!!)

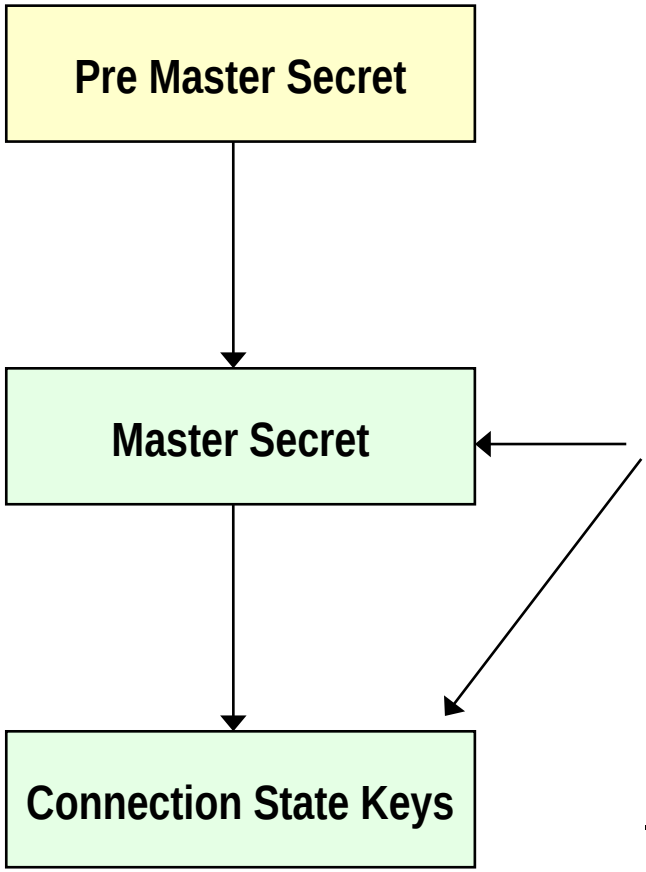


Why a separate protocol, and not part of the Handshake?

Wise choice! TLS specification allows aggregation of multiple messages of a SAME upper layer protocol into single TLS Record. How this possible with a Change Cipher Spec? (TLS Record is either ALL encrypted, or NOT encrypted)

TLS Key Computation

Secret hierarchy



Exchanged (RSA) or computed (DH) during handshake

- could be always the same value for same C-S pair (e.g., in the case fo fixed DH)

Master Secret computed from:

- Pre master
- random values from C & S
- timestamps from C & S

→ Recomputed at session resumption

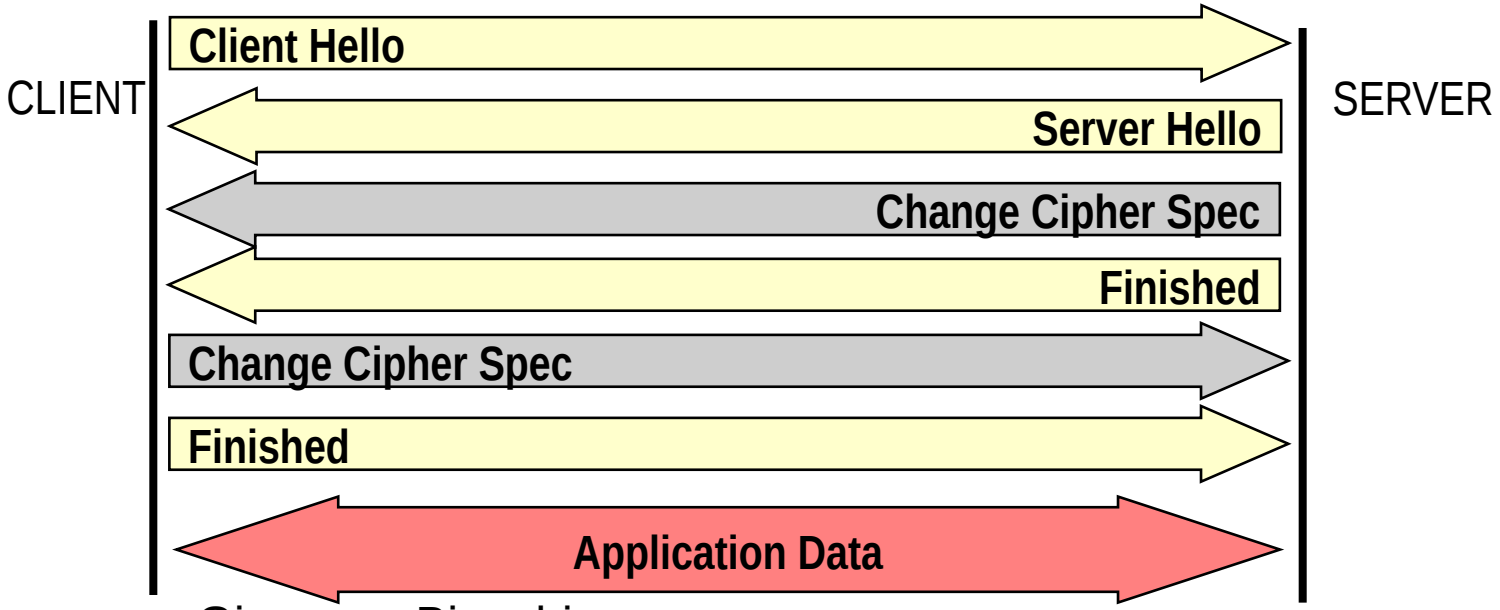
Up to 6 keys:

- encryption keys (write/read = C/S)
- authentication keys (write/read = C/S)
- initialization vector if needed (write/read)

→ Recomputed at session resumption

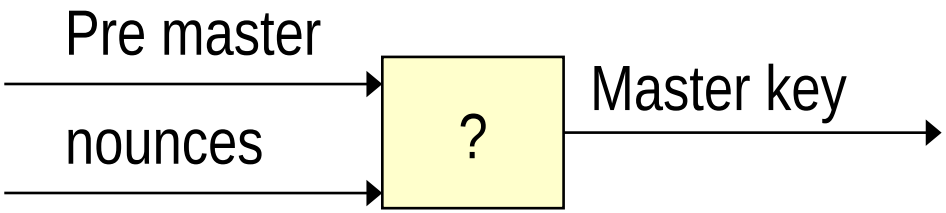
Abbreviated handshake (3-way) (session resumption)

- **Used to re-generate key material**
  - For new connection
- **Avoids to reauthenticate peers**
  - Done at start of session only
- **Avoid to re-exchange pre-master-secret**
- **Exchange new TS+random values**

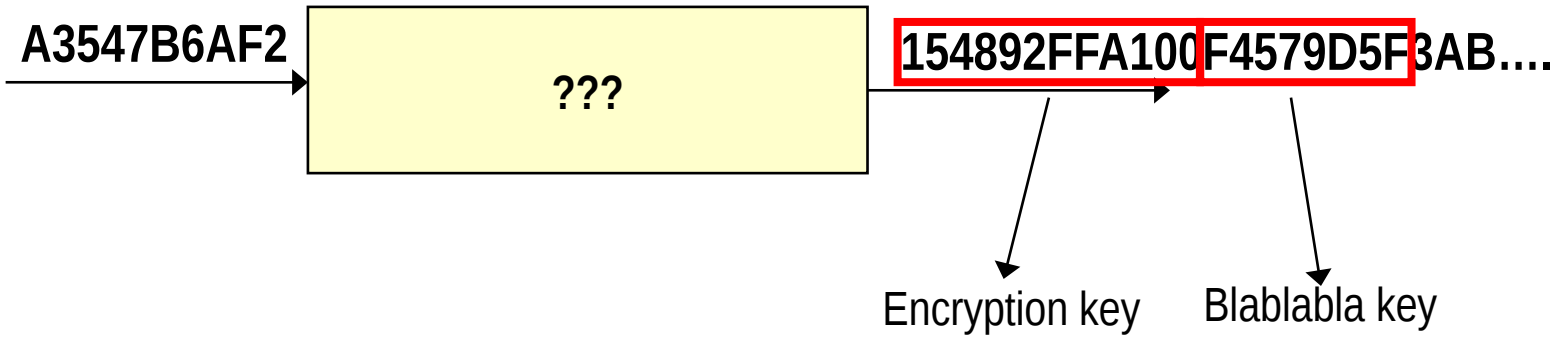


Initial keying, re-keying: Extract-then-expand

1) "**Extract**": Add randomness in key generation

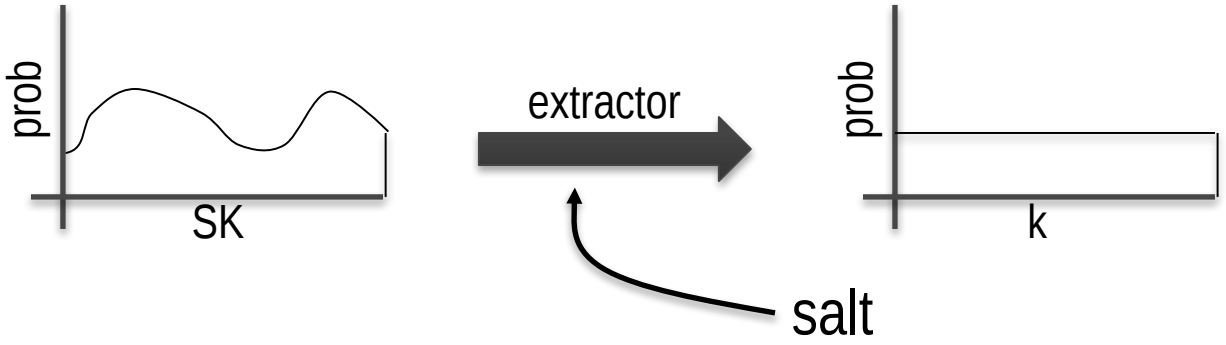


2) "**Expand**" initially limited secret to the needed amount of crypto material



Extract-then-Expand paradigm

Step 1: extract pseudo-random key k from source key SK



salt: a fixed non-secret string chosen at random

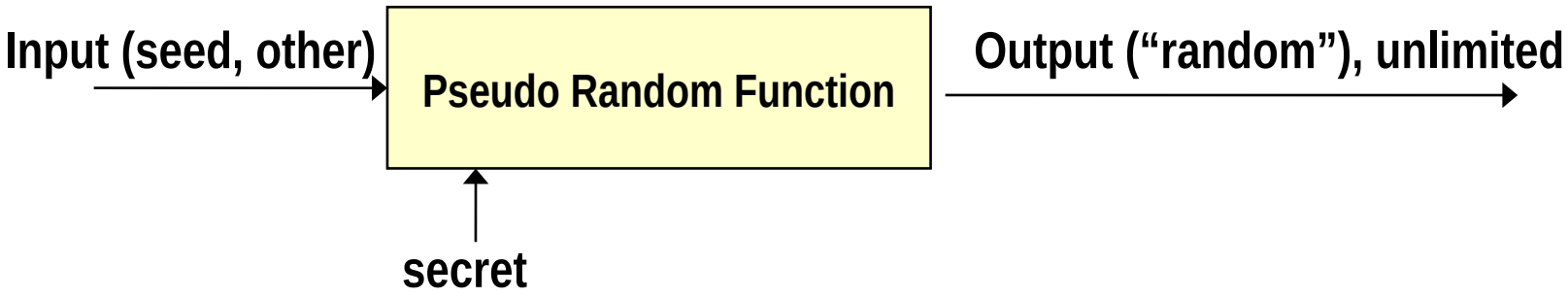
step 2: expand k by using it as a PRF key

Why extract?

- **Constructions rely on the assumption that key k is random**
  - Random = uniform distribution!
- **What if this is not true?**
  - Biases from HW random generator
  - Fixed DH exchange (!)

- **Extract:**
  - guarantees random master key
  - "salts" it with nonces

Basic building block: PRF



- **Fundamental "brick":**
  - GOOD (and possibly fast!) PRF
- **PRFs in TLS**
  - SSL 3.0: not fully satisfactory PRF
  - TLS 1.0 and 1.1: good PRF but… hard-coded!
    - Construction used MD5 and SHA-1 (now weak)
    - *(unlike IPsec) TLS forgot that "good" in crypto is not forever….*
  - **TLS 1.2:negotiable PRF (at last ☺)**
    - **default one based on SHA-256**

PRF from a 1-way hash: Expansion function P_{hash}

- → **A0 = seed**
- → **A1 = $HMAC_{hash}$(A0)**
- → **A2 = $HMAC_{hash}$(A1)**
- → **A3 = $HMAC_{hash}$(A2)**
- → **…**

Same secret used in all HMACs
Hash = chosen hash function

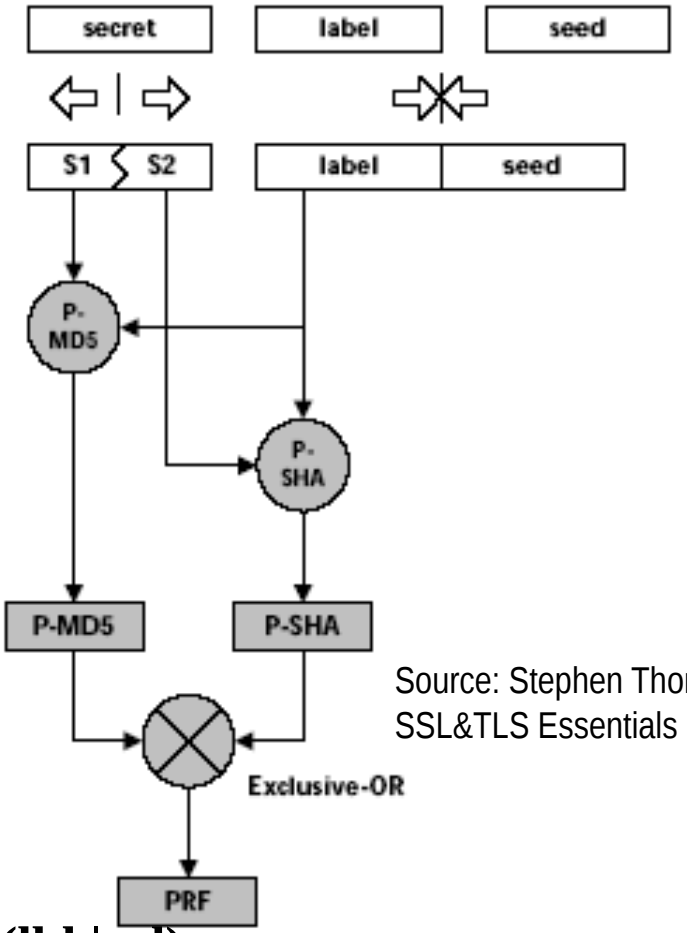
P_{hash} = | $HMAC_{hash}(A_1 \mid seed)$ | $HMAC_{hash}(A_2 \mid seed)$ | $HMAC_{hash}(A_3 \mid seed)$ | ……

- → **P_{hash} function of:**
  - ⇨ Chosen hash function
  - ⇨ Secret
  - ⇨ seed
- → **P_{hash} size: any (unlike HMAC size)**

PRF until TLS 1.1: hard coded!

- **Employs two hash algorithms**
  - Idea: combine both MD5 and SHA-1
  - Greater robustness! Must break both…
    - Indeed ☺

- **Input:**
  - Desired length
  - Secret
    - Assume even size, split it in two parts (S_1,S_2)
  - Seed
  - Label (an ascii string)



Source: Stephen Thomas, SSL&TLS Essentials

- **PRF(scrt, lbl, sd) = P_{MD5}(lbl | sd) XOR P_{SHA-1}(lbl | sd)**
  - MD5 uses S1 as secret
  - SHA-1 uses S2 as secret

TLS 1.2: PRF from P_{hash}

PRF(secret, label, seed) = P_{hash}(label|seed)

- **TLS1.2:**
  - Default: this construction with SHA-256
    - 32 bytes hash
  - But also negotiable PRF
    - Hence possibly very different and not based on HMAC

- **Example: Master secret generation**
  - Inputs:
    - Secret = Pre-master-secret
    - Seed = Nonces (client-random & server-random)
  - Further input:
    - Label = string “master secret”
  - Computation: first 48 bytes of:
    - **PRF(pre-master-secret, “master secret”, Client-Random | Server-Random)**

Key generation

- **Master secret [48 bytes]:**
  - PRF(pre-master-secret, “master secret”, Client-Random | Server-Random)
- **Key Block [size depends on cipher suites]:**
  - PRF(master-secret, “key expansion”, Server-Random | Client-Random)
- **Individual keys:**
  - Partition key block into up to 6 fields in the following order:
  - Client MAC, Server MAC, Client Key, Server Key, Client IV, Server IV

PRF used also in computation of finished message instead of “normal” MD5 or SHA-1 Hash. E.g. For client finished message:

PRF(master-secret, “client finished”, MD5(all handshake msg) | SHA-1(all handshake msg)) [12 bytes]

PRF in TLSv1.3: HKDF

→ HMAC-based Key Derivation Function

- ⇨ H. Krawczyk 2010, provably secure
- ⇨ New name: KDF → precisely states what this tool is used for
- ⇨ Standardized: RFC 5869
- ⇨ Included in newer TLS version

- Start from master key K (extracted from HMAC)
- use context string (e.g. TLS' label | nonces) to distinguish (eventual) multiple usage of K

HKDF= | $HMAC_{hash, K}$(context string\|0) | $HMAC_{hash, K}$(context string\|1) | …